



Fast Authentication between flying UAVs

I have taken Project instead of course in this semester under a professor, sir told me to do a research on topic, sir have given me different topics and i choose Fast Authentication between flying UAVs in which i have design a technology two authenticate between flying UAVs in milliseconds and share information and data securely and very fast, it has many real life purpose which you can think of so this is the project i am trying to choose, here in this i have design everything from scratch to algorithm to implementation in 4 months help me to and give few ideas where to start and how to do, what to read and research in details

I'll help you design a comprehensive research roadmap for your Fast Authentication between Flying UAVs project. Let me gather all the essential information and create a detailed guide for your 4-month journey.

Fast Authentication Between Flying UAVs: A Comprehensive Research & Implementation Guide

Executive Overview

Your project on **Fast Authentication between Flying UAVs** addresses one of the most critical challenges in modern drone operations: establishing secure, millisecond-level authentication for UAV-to-UAV (U2U) communication. This research is timely and impactful, with applications spanning military coordination, disaster response, autonomous delivery systems, swarm operations, and civilian infrastructure monitoring. ^{[1] [2] [3]}

Current research demonstrates that authentication latency in UAV networks can be reduced to **sub-100 millisecond** range, with some implementations achieving **8-50 milliseconds** for complete authentication cycles. Your goal of millisecond-level authentication is both ambitious and achievable with the right combination of cryptographic techniques and protocol design. ^{[2] [3] [4] [5]}

Understanding the Problem Space

Why Fast Authentication Matters

Flying UAVs operate in highly dynamic environments where:

- **Real-time coordination** is essential for collision avoidance and formation flying ^{[6] [7]}
- **Swarm operations** require continuous authentication of multiple drones simultaneously ^{[3] [8]}
- **Mission-critical applications** demand latencies under 100ms for effective response ^{[9] [10]}

- **Security threats** including GPS spoofing, man-in-the-middle attacks, and drone hijacking are pervasive^{[11] [2] [3]}

Traditional authentication methods (PKI, digital certificates) are too **resource-intensive** for battery-constrained UAVs and too **slow** for real-time operations. Your research must balance three critical factors:^{[12] [13]}

1. **Speed** (millisecond-level latency)
2. **Security** (resistance to attacks)
3. **Resource efficiency** (low computational overhead for constrained devices)

Current State-of-the-Art

Recent research has achieved impressive authentication performance:^{[13] [4] [14] [5] [2] [3]}

Protocol/Method	Authentication Time	Key Technology
Proxy Signature D2D	8.36 ms	Proxy signatures + 5G ^[5]
PUF-based Switch Authentication	214 μ s (microseconds)	Hardware offloading to P4 switches ^[4]
SENTINEL Protocol	3.1 $\times$ faster than TLS-IoT	HMAC-SHA256 + session keys ^[13]
HOOPOE Protocol	Sub-100 ms handover	AES-RSA hybrid + OMNeT++ simulation ^[3]
SELTHA Protocol	Reduced computational overhead	TOTP + hashed authentication ^[12]
Multilayer Authentication	3.255s latency, 90.15% throughput	Hierarchical approach ^{[15] [16]}

Core Technologies You Should Master

1. Physical Unclonable Functions (PUFs)

PUFs are emerging as the **gold standard** for UAV authentication.^{[17] [18] [4] [14] [19] [20]}

Why PUFs are ideal:

- Generate unique, unclonable device "fingerprints" from manufacturing variations^{[21] [20] [22]}
- Enable **on-the-fly key generation** without storing secrets in memory^{[20] [22]}
- Provide hardware-based root of trust resistant to cloning attacks^{[23] [20]}
- Achieve authentication in **sub-millisecond** timeframes when hardware-accelerated^[4]

Key concepts to study:

- Challenge-Response Pairs (CRPs)^{[14] [19] [20]}
- PUF types: SRAM PUF, Ring Oscillator PUF, Arbiter PUF^{[21] [20]}

- Integration with lightweight hash functions (SHA-256, SPONGENT-128) [\[24\]](#) [\[14\]](#)

Implementation approach:

- Use PUF for device identification during enrollment phase
- Store CRPs securely at Ground Control Station (GCS) [\[4\]](#) [\[23\]](#)
- During authentication, GCS sends challenge; UAV responds with PUF-generated response
- Verify response matches stored CRP [\[14\]](#) [\[23\]](#)

2. Lightweight Cryptography

For resource-constrained UAVs, lightweight algorithms are essential. [\[25\]](#) [\[26\]](#) [\[27\]](#) [\[28\]](#)

NIST-standardized lightweight algorithms:

Algorithm	Use Case	Performance
ASCON-128/128a	AEAD encryption	Optimal for UAVs; low latency [25] [26]
GIFT-COFB	Block cipher	Hardware-friendly [26]
Elephant	AEAD	Low memory footprint [26]
Xoodyak	Hashing + encryption	Compact implementation [26]

Traditional lightweight options:

- **ChaCha20-Poly1305:** Fast authenticated encryption, used for flight data [\[27\]](#)
- **AES-256-GCM:** Industry standard, but higher overhead [\[29\]](#) [\[30\]](#)

Elliptic Curve Cryptography (ECC):

- **ECC-256/512:** Provides strong security with smaller key sizes than RSA [\[31\]](#) [\[30\]](#) [\[32\]](#) [\[33\]](#)
- **ECDH** (Elliptic Curve Diffie-Hellman): For key exchange [\[30\]](#) [\[34\]](#)
- Suitable for UAVs with moderate processing capability [\[30\]](#) [\[14\]](#)

3. Hash-Based Message Authentication Codes (HMAC)

- **HMAC-SHA256:** Fast, widely supported, used in SENTINEL protocol [\[35\]](#) [\[13\]](#)
- **HMAC with session keys:** Enables rapid verification without heavy cryptography [\[13\]](#)
- Can authenticate messages in **microseconds** once session established [\[35\]](#) [\[13\]](#)

4. Blockchain & Distributed Ledger (Optional Advanced Feature)

For swarm scenarios with no central authority: [\[18\]](#) [\[36\]](#) [\[11\]](#) [\[17\]](#)

- Store authentication credentials immutably [\[11\]](#) [\[17\]](#)
- Enable trustless verification between UAVs [\[36\]](#) [\[18\]](#)
- **Challenge:** High latency and overhead; requires optimization for real-time use [\[8\]](#) [\[17\]](#)

Research Methodology: Your 4-Month Roadmap

Month 1: Foundation & Literature Review (Weeks 1-4)

Week 1-2: Problem Definition & Core Literature

Key activities:

- Define your **specific research questions**:
 - What authentication latency can you achieve?
 - Which cryptographic primitives provide best speed/security tradeoff?
 - How does your protocol scale to swarm scenarios (5, 10, 50 UAVs)?

Essential papers to read (priority order):

1. **Accelerating PUF-based UAV Authentication Protocols** (Ranjitha et al., 2019) - Demonstrates 214 μ s authentication using P4 switches^[4]
2. **HOOPOE: High Performance Anonymous Handover Authentication** (IEEE 2023) - Fast handover for out-of-zone UAVs^[3]
3. **SENTINEL: Secure and Efficient Authentication Framework** (MDPI 2020) - 3.1 $\times$ faster than TLS^[13]
4. **Efficient Authentication of Drones to mmWave Mesh Networks** (IEEE 2020) - Proxy signatures for post-disaster scenarios^[2]
5. **A Proxy Signature-Based Drone Authentication in 5G D2D Networks** (2021) - 8.36ms authentication time^[5]
6. **PUF secured lightweight mutual authentication protocol** (Chandran et al., 2024) - Multi-UAV networks with PUF^[14]
7. **Light-Weight Cryptography Algorithms for UAV-Networks** (2025) - ASCON performance analysis^[25]
8. **Lightweight Cryptography in UAV systems** (Skorobahatko et al., 2025) - Practical FPGA experiments^[26]

Week 3-4: Technology Deep Dive

Focus areas:

- Study **PUF implementation** details and CRP management^{[20] [4] [14]}
- Analyze **lightweight crypto** algorithms (ASCON, ChaCha20, ECC)^{[26] [27] [25]}
- Understand **wireless communication protocols** for UAVs (MQTT, UDP, MAVLink)^{[37] [38] [39]}
- Review **attack vectors**: replay attacks, man-in-the-middle, spoofing, cloning^{[40] [41] [2] [3]}

Deliverable:

- Literature review document (8-10 pages)

- Comparative table of existing authentication protocols
- Identification of research gap your project will address

Month 2: System Design & Algorithm Development (Weeks 5-8)

Week 5-6: Architecture Design

Design your authentication protocol with these components:

- 1. Enrollment Phase** (one-time, offline)
 - Generate PUF CRPs for each UAV
 - Store CRPs securely at GCS or in blockchain
 - Assign unique UAV identifiers
- 2. Authentication Phase** (real-time, online)
 - Lightweight handshake (2-3 message exchanges)
 - Challenge-response using PUF or HMAC
 - Session key establishment using ECDH or pre-shared keys
- 3. Secure Communication Phase**
 - Encrypt data using ASCON-128a or ChaCha20-Poly1305
 - Message authentication using HMAC-SHA256
 - Periodic re-authentication (every 30-60 seconds)

Architecture diagram elements:

- UAV nodes (with PUF hardware or simulated PUF)
- Ground Control Station (GCS) or leader UAV (in swarm)
- Communication channels (wireless ad-hoc network)
- Authentication server/blockchain (if applicable)

Week 7-8: Algorithm Specification

Develop detailed algorithms for:

Algorithm 1: UAV Registration

```
Input: UAV_ID, PUF_Hardware
Output: Stored CRPs at GCS
1. Generate N challenge-response pairs using PUF
2. Store CRPs in secure database with UAV_ID
3. Issue authentication certificate to UAV
```

Algorithm 2: Fast Mutual Authentication

Input: UAV_A, UAV_B (or GCS)
Output: Authenticated session, session_key

1. UAV_A $\rightarrow$ UAV_B: {UAV_ID_A, Nonce_A, Timestamp}
2. UAV_B verifies Timestamp (prevent replay)
3. UAV_B retrieves Challenge_A from CRP database
4. UAV_B $\rightarrow$ UAV_A: {Challenge_A, Nonce_B}
5. UAV_A computes Response_A = PUF(Challenge_A)
6. UAV_A $\rightarrow$ UAV_B: {Response_A, HMAC(Response_A || Nonce_B)}
7. UAV_B verifies Response_A matches stored CRP
8. If valid, compute session_key = ECDH(Private_B, Public_A) XOR Response_A
9. Both UAVs now share session_key for encrypted communication

Algorithm 3: Secure Data Exchange

Input: Message M, session_key
Output: Encrypted message C

1. Encrypt: C = ASCON_Encrypt(M, session_key, Nonce)
2. MAC = HMAC-SHA256(C, session_key)
3. Transmit: {C, MAC, Nonce}
4. Receiver verifies MAC, decrypts C

Deliverable:

- System architecture diagram
- Detailed protocol specification (pseudo-code)
- Security analysis: resistance to common attacks^{[40] [2] [3]}

Month 3: Implementation & Development (Weeks 9-12)

Week 9-10: Simulation Environment Setup

Choose your simulation platform:

Option A: NS-3 + Gazebo + ROS (Most Realistic)^{[42] [43] [44] [45] [46] [37]}

- **NS-3:** Network simulation (latency, packet loss, throughput)
- **Gazebo:** 3D UAV physics simulation
- **ROS:** Robot Operating System for UAV control
- **Integration:** Use middleware like CORNET, DroNS-3, or FANS^{[45] [46] [37]}

Option B: MATLAB/Simulink + UAV Toolbox^{[47] [48] [49] [50]}

- **UAV Toolbox:** Built-in drone dynamics and sensors
- **Simulink:** Visual programming for control systems
- **SwarmLab:** MATLAB-based swarm simulator^[48]

- Easier to prototype, good for algorithm validation

Option C: Python + Custom Simulation^[38] ^[51] ^[52]

- **Lightweight** and flexible
- Use **AirSim + ROS wrapper** for realistic environments^[51]
- Ideal if you're comfortable with Python

My recommendation: Start with **Python + Custom Simulation** for rapid prototyping of authentication protocol, then validate with **NS-3 + Gazebo** for realistic network conditions.

Week 11-12: Core Implementation

Implement in phases:

Phase 1: Cryptographic Primitives

- Implement or use existing libraries:
 - **PyCryptodome** (Python): HMAC, SHA-256, AES
 - **Ascon** library: Lightweight AEAD^[25]
 - **ECC** library: `ecdsa` or `cryptography` package
- Test individual crypto operations, measure latency

Phase 2: Authentication Protocol

- Implement your 3-phase protocol (enrollment, authentication, communication)
- Simulate UAV-to-UAV and UAV-to-GCS scenarios
- Use **UDP sockets** for low-latency communication^[53] ^[2]

Phase 3: Network Simulation

- Set up **2-5 UAVs** in simulated environment
- Implement wireless channel (packet loss, delay, bandwidth limits)
- Test authentication under various conditions (interference, mobility)

Phase 4: Performance Measurement

- Measure:
 - **Authentication latency** (end-to-end time)
 - **Computational overhead** (CPU usage, memory)
 - **Communication overhead** (bytes transmitted)
 - **Throughput** (successful authentications per second)
 - **Packet Delivery Ratio** (PDR)

Deliverable:

- Working prototype with 2-5 simulated UAVs
- Performance benchmarks comparing your protocol to baseline (e.g., TLS, simple PSK)

Month 4: Testing, Evaluation & Documentation (Weeks 13-16)

Week 13: Comprehensive Testing

Test scenarios:

1. **Normal operation:** 2 UAVs, optimal conditions
2. **Scalability:** 5, 10, 20 UAVs (measure latency degradation)
3. **Mobility:** UAVs moving at 10-20 m/s^{[7] [6]}
4. **Packet loss:** 5%, 10%, 20% loss rates^{[15] [16]}
5. **Attack scenarios:**
 - Replay attack (send old authentication messages)
 - Man-in-the-middle (intercept and modify messages)
 - Impersonation (fake UAV tries to authenticate)
 - Denial-of-Service (flood authentication requests)

Week 14: Performance Optimization

If initial results don't meet targets, optimize:

- **Reduce message exchanges:** Can you authenticate in 2 messages instead of 3?
- **Pre-compute:** Generate session keys in advance when possible
- **Batch authentication:** Authenticate multiple UAVs simultaneously^[23]
- **Hardware acceleration:** Offload crypto to dedicated chip (if using actual hardware)

Week 15: Documentation & Paper Writing

Structure your research paper:

1. **Abstract** (150-200 words)
 - Problem statement
 - Your solution approach
 - Key results (e.g., "achieved 15ms authentication")
2. **Introduction** (2-3 pages)
 - Motivation: Why fast UAV authentication matters
 - Challenges: Speed vs. security tradeoff
 - Contribution: Your novel approach
3. **Related Work** (3-4 pages)
 - Survey of existing protocols^{[12] [2] [3] [13] [4] [14]}
 - Comparison table highlighting gaps
4. **System Model & Threat Model** (2 pages)

- Network architecture
- Adversary capabilities (Dolev-Yao model) ^[14]
- Security requirements

5. **Proposed Protocol** (4-5 pages)

- Detailed description of your 3 phases
- Algorithms with pseudo-code
- Security analysis (how you prevent each attack)

6. **Implementation & Evaluation** (4-5 pages)

- Simulation setup (tools, parameters)
- Performance metrics
- Results with graphs:
 - Authentication latency (bar chart or CDF) ^[4]
 - Scalability (latency vs. number of UAVs)
 - Comparison with baseline protocols

7. **Discussion** (1-2 pages)

- Limitations
- Real-world deployment considerations
- Future work

8. **Conclusion** (1 page)

Week 16: Final Presentation Preparation

Create a compelling 15-20 minute presentation:

- **Slide 1:** Title & motivation with real-world scenarios
- **Slides 2-3:** Problem statement & challenges
- **Slides 4-5:** Related work & research gap
- **Slides 6-8:** Your protocol design (architecture diagrams, flowcharts)
- **Slides 9-11:** Implementation & results (graphs, performance tables)
- **Slide 12:** Demo video (if possible, show UAVs authenticating in simulation)
- **Slide 13:** Contributions & future work
- **Slide 14:** Q&A

Deliverable:

- Complete research paper (12-20 pages)
- Presentation slides
- Source code (GitHub repository) ^{[54] [55] [56] [57]}
- Technical documentation

Implementation Tools & Resources

Software Frameworks

Tool	Purpose	Learning Curve
NS-3 [42] [43] [45]	Network simulation	Moderate-High
Gazebo [44] [58] [59] [50]	3D UAV physics	Moderate
ROS/ROS2 [44] [46] [37] [39]	Robotics middleware	Moderate
MATLAB UAV Toolbox [47] [49] [50]	UAV simulation	Low-Moderate
Python + AirSim [51]	Visual simulation	Low
OMNeT++ [3] [50]	Network simulation	Moderate

Cryptographic Libraries

Library	Language	Features
PyCryptodome	Python	AES, HMAC, SHA, ECC
Crypto++	C++	Comprehensive crypto toolkit
libsodium	C	Modern, easy-to-use crypto
ASCON implementation [25]	C/Python	Lightweight AEAD
mbedTLS	C	Embedded/IoT crypto

UAV Platforms & Controllers

- **PX4 Autopilot** - Open-source flight control [\[43\]](#) [\[60\]](#) [\[49\]](#)
- **ArduPilot** - Mature UAV software [\[61\]](#) [\[60\]](#) [\[45\]](#)
- **MAVLink** - Lightweight messaging protocol [\[58\]](#) [\[37\]](#) [\[38\]](#)
- **QGroundControl** - Ground control station software [\[60\]](#)

Development Hardware (if available)

- **Pixhawk 4** - Flight controller for testing [\[62\]](#) [\[37\]](#) [\[60\]](#)
- **Raspberry Pi 4 + Navio2** - Companion computer [\[38\]](#)
- **NVIDIA Jetson Nano** - For AI/ML-enhanced security [\[37\]](#)
- **Xbee modules** - Wireless communication [\[63\]](#)

Real-World Applications & Impact

Your fast authentication protocol can enable:

1. Disaster Response & Search-and-Rescue^{[64] [65] [66]}

- Multiple UAVs coordinate to map disaster areas
- Fast authentication ensures only authorized rescue drones operate
- Sub-100ms latency enables real-time victim detection and coordination

2. Military & Defense Operations^{[67] [6] [2]}

- Swarm tactics require instant friend-or-foe identification^[10]
- Secure communication prevents adversary infiltration
- Millisecond latency critical for coordinated strikes

3. Autonomous Package Delivery^{[68] [1]}

- UAVs authenticate with delivery hubs and customer locations
- Fast authentication reduces wait time, improves efficiency
- Security prevents package theft via drone hijacking

4. Smart Agriculture & Infrastructure Monitoring^[64]

- Fleets of UAVs monitor crops, pipelines, power lines
- Authentication ensures data integrity
- Low latency enables real-time anomaly response

5. Urban Air Mobility (UAM)^{[69] [43]}

- Air taxis and passenger drones require stringent safety
- Fast authentication for collision avoidance systems
- Integration with 5G networks for ultra-low latency^{[70] [69]}

Success Metrics & Target Performance

Based on current research, aim for these benchmarks:

Metric	Target	Excellent	Baseline (for comparison)
Authentication Latency	< 50 ms	< 20 ms	TLS: ~500 ms ^[13]
Throughput	> 80%	> 90%	Traditional: 60-70% ^[15]
Packet Delivery Ratio	> 85%	> 95%	Standard: 70-80% ^[37]
Computational Overhead	< 10% CPU	< 5% CPU	RSA/PKI: 20-30% ^[12]

Metric	Target	Excellent	Baseline (for comparison)
Communication Overhead	< 1 KB/auth	< 500 bytes	Digital certs: 2-4 KB ^[2]
Scalability	10+ UAVs	50+ UAVs	N/A
Energy Consumption	< 50 mW	< 20 mW	Heavy crypto: 100+ mW ^[71]

Key Research Contributions You Can Make

To ensure your project is publishable and impactful, focus on **at least 2-3** of these novel contributions:

1. **Hybrid PUF + Lightweight Crypto Protocol:** Combine PUF-based device authentication with ASCON encryption for optimal speed-security balance ^{[25] [4] [14]}
2. **Batch Authentication for Swarms:** Design a protocol that authenticates multiple UAVs simultaneously in $O(\log n)$ or $O(1)$ time rather than $O(n)$ ^[23]
3. **Predictive Authentication:** Use mobility prediction (trajectory, flight plan) to pre-authenticate UAVs before they enter communication range ^[72]
4. **Context-Aware Authentication:** Adjust authentication rigor based on threat level - lighter authentication in safe zones, stronger in hostile areas ^[53]
5. **5G/6G Integration:** Design protocol specifically for URLLC (Ultra-Reliable Low-Latency Communication) with sub-1ms authentication ^{[69] [10] [70]}
6. **Hardware-Accelerated Crypto:** Implement authentication logic on FPGA or P4-programmable switches for <1ms latency ^[4]
7. **Quantum-Resistant UAV Authentication:** Future-proof by using post-quantum cryptography (CRYSTALS-Kyber) ^{[73] [74]}

Common Pitfalls to Avoid

1. **Don't simulate with perfect conditions only:** Test with packet loss, interference, mobility ^{[16] [15] [53]}
2. **Don't ignore computational constraints:** UAVs have limited CPU/battery; measure real overhead ^{[12] [26] [25]}
3. **Don't use synthetic/fake data:** If presenting performance numbers, they must come from actual simulation or hardware tests ^[50]
4. **Don't overlook attack scenarios:** Your security analysis must address replay, MITM, DoS, cloning ^{[2] [3] [40] [14]}
5. **Don't claim novelty without literature review:** Thoroughly compare your work to existing protocols ^{[3] [2] [12] [13] [14] [4]}
6. **Don't build everything from scratch:** Leverage existing libraries (PyCryptodome, NS-3, ROS) - focus on your novel protocol design ^{[45] [50] [37]}

Recommended Reading Order (Core Papers)

Week 1: Foundation

1. Accelerating PUF-based UAV Authentication (Ranjitha et al.)^[4]
2. SENTINEL Framework (MDPI 2020)^[13]
3. Efficient Authentication in mmWave Networks (IEEE 2020)^[2]

Week 2: Advanced Techniques

4. HOOPOE Handover Authentication (IEEE 2023)^[3]
5. PUF Secured Lightweight Protocol (Chandran et al., 2024)^[14]
6. Light-Weight Cryptography for UAV-Networks (Patel et al., 2025)^[25]

Week 3: Security & Attacks

7. Two Secure Authentication Protocols (Nature 2025)^[40]
8. Towards Secure Wireless Mesh Networks for UAV Swarms (IEEE 2021)^[7]
9. Survey on Security of UAV Swarm Networks (ACM 2024)^[75]

Week 4: Implementation Insights

10. Proxy Signature-Based Drone Authentication in 5G (2021)^[5]
11. Low-Latency Communication Protocols for Drone IFF (Decent Cybersecurity 2024)^[10]
12. Scalable Mutual Authentication Protocol (Bansal et al.)^[23]

GitHub Resources & Open-Source Projects

Explore these repositories for reference implementations: ^[55] ^[56] ^[57] ^[76] ^[54]

1. **SecAuthUAV** - Study of UAV authentication protocols^[54]
2. **UAV-EXPRESS** - Real-world UAV delivery system framework^[55]
3. **Autonomous-UAVs** - List of open-source algorithms and resources^[56]
4. **UAVs-Dataset-Under-Cyberattacks** - Dataset with DoS, replay, MITM attacks^[57]
5. **asv-drones** - Modular UAS application with authentication examples^[76]

Final Recommendations & Next Steps

Immediate Actions (This Week):

1. **Set up your development environment:**
 - Install Python 3.8+ with PyCryptodome, numpy, matplotlib
 - OR install MATLAB with UAV Toolbox
 - OR install NS-3 + Gazebo + ROS (if you're ambitious)

2. **Create project structure:**

```
UAV-FastAuth/  
├── literature/          # Store PDFs of papers
```

```

├── design/           # Architecture diagrams, flowcharts
├── src/              # Source code
│   ├── crypto/       # Cryptographic primitives
│   ├── protocol/     # Authentication protocol
│   └── simulation/    # Network simulation
├── results/         # Performance data, graphs
├── docs/            # Documentation
└── README.md

```

3. Start literature review:

- Download and read the 12 core papers listed above
- Create a comparison table of existing protocols
- Identify your unique contribution

Success Mindset:

- This is **ambitious but achievable** in 4 months with focused effort
- **Iterate quickly:** Build simple version first, then optimize
- **Document as you go:** Don't leave documentation to the end
- **Meet with your professor weekly:** Get feedback early and often
- **Join UAV/drone communities:** Reddit r/diydrones, ROS Discourse, Stack Overflow

Your project has real potential to contribute to UAV security research. With sub-100ms authentication, you'll be competitive with state-of-the-art protocols. Stay focused, test rigorously, and document everything. Good luck! ☺



1. <https://arxiv.org/abs/2501.04480>
2. <https://ieeexplore.ieee.org/document/9322282/>
3. <https://ieeexplore.ieee.org/document/10082875/>
4. <https://praveenabt.github.io/publications/puf-uav-p4.pdf>
5. https://inl.digitallibrary.inl.gov/sites/sti/sti/Sort_28987.pdf
6. <https://ieeexplore.ieee.org/document/10430396/>
7. <https://ieeexplore.ieee.org/document/9599886/>
8. <https://www.techscience.com/cmc/online/detail/25522>
9. <https://ieeexplore.ieee.org/document/10683537/>
10. <https://decentcybersecurity.eu/low-latency-communication-protocols-for-drone-iff-ensuring-swift-and-secure-identification/>
11. <https://ieeexplore.ieee.org/document/9861105/>
12. <https://ieeexplore.ieee.org/document/10178552/>
13. <https://www.mdpi.com/2076-3417/10/9/3149/pdf>
14. <https://www.sciencedirect.com/science/article/abs/pii/S1389128624005498>
15. <https://www.mdpi.com/1424-8220/24/3/996/pdf?version=1707032657>

16. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10857715/>
17. <https://www.arxiv.org/pdf/2402.15817.pdf>
18. <http://arxiv.org/pdf/2407.00630.pdf>
19. <https://www.sciencedirect.com/science/article/abs/pii/S0045790623000873>
20. <https://arxiv.org/html/2512.21827v1>
21. <https://promwad.com/news/physical-unclonable-functions-electronics-security>
22. <https://www.synopsys.com/glossary/what-is-a-physical-unclonable-function.html>
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26. <https://tacs.ipt.kpi.ua/article/download/326898/325720>
27. https://thesai.org/Downloads/Volume16No5/Paper_27-Securing_UAV_Flight_Data_Using_Lightweight_Cryptography.pdf
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33. <https://onlinelibrary.wiley.com/doi/10.1155/2024/5808088>
34. <https://www.projectguideline.com/implementation-of-elliptic-curve-cryptography-in-ns-2/>
35. <http://arxiv.org/pdf/2410.09085.pdf>
36. <https://downloads.hindawi.com/journals/scn/2022/2330052.pdf>
37. <https://dl.acm.org/doi/10.1145/3747204.3747223>
38. <https://www.ijert.org/research/pi-drone-using-python-IJERTV10IS090105.pdf>
39. <https://dronecode.org/projects/>
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